Deep Autoencoders With Multitask Learning for Bilinear Hyperspectral Unmixing

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***Abstract*— Hyperspectral unmixing is an important problem for remotely sensed data interpretation. It amounts at estimating the spectral signatures of the pure spectral constituents in the scene (endmembers) and their corresponding subpixel fractional abundances. Although the unmixing problem is inherently nonlinear (due to multiple scattering), the nonlinear unmixing of hyperspectral data has been a very challenging problem. This is because nonlinear models require detailed knowledge about the physical interactions between the sunlight scattered by multiple materials. In turn, bilinear mixture models (BMMs) can reach good accuracy with a relatively simple model for scattering. In this article, we develop a new BMM and a corresponding unsupervised unmixing approach which consists of two main steps. In the first step, a deep autoencoder is used to linearly estimate the endmember signatures and their associated abundance fractions. The second step refines the initial (linear) estimates using a bilinear model, in which another deep autoencoder (with a low-rank assumption) is adapted to model second-order scattering interactions. It should be noted that in our developed BMM model, the two deep autoencoders are trained in a mutually interdependent manner under the multitask learning framework, and the relative reconstruction error is used as the stopping criterion. The effectiveness of the proposed method is evaluated using both synthetic and real hyperspectral data sets. Our experimental results indicate that the proposed approach can reasonably estimate the nature of nonlinear interactions in real scenarios. Compared with other state-of-the-art unmixing algorithms, the proposed approach demonstrates very competitive performance.**

***Index Terms*— Autoencoder, bilinear mixture, deep learning, hyperspectral nonlinear unmixing, multitask learning (MTL).**

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I. Introduction

HYPERSPECTRAL remote sensing, which combines traditional imaging and physical (spectral) analysis techniques, has been widely used for Earth observation purposes [1]. Hyperspectral sensors can acquire fine spectral information and rich spatial details of the target scenes, thus achieving strong discrimination ability for different land-cover classes and promoting the development of quantitative remote sensing [2], [3]. Nowadays, hyperspectral imagery has been readily used in many geoscience fields, including environmental monitoring, precision agriculture, and land-use surveying [4], [5]. However, due to the relatively low spatial resolution of hyperspectral images, plenty of pixels are mixed by several different substances, leading to inaccuracies in the understanding and quantification of the considered scenes [6]. To address this issue, spectral unmixing (SU) has been adopted to decompose each pixel spectrum into a collection of pure constituent spectra (called endmembers), and their corresponding abundance fractions [7]. Most SU approaches are based on linear or nonlinear spectral mixture models [8]–[10].

The linear mixture model (LMM) assumes that the endmembers interact linearly with the incident radiation at a subpixel level [11]. Many approaches have been developed under the LMM assumption. These algorithms generally perform unmixing without considering the scattering interactions between materials. Among many examples, we can mention N-FINDR [12], vertex component analysis (VCA) [13], minimum volume simplex analysis [14], joint Bayesian algorithm [15], incremental proximal sparse and low-rank unmixing [16], SU with low-rank attribute [17], robust collaborative non-negative matrix factorization (NMF) [18], $L_{1/2}$-NMF [19], and sparsity-constrained deep NMF [20].

In recent years, autoencoders have also been widely used for developing robust SU approaches based on the LMM.

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Autoencoders, as a learning model based on artificial neural networks, provide a combination of an encoder and a decoder to carry out unsupervised learning [21]. The encoder transforms the input data into a code, and the decoder reconstructs the input data from the corresponding code [22]. By means of a shared weight strategy between adjacent layers (as well as contrastive divergence), autoencoders can be trained quite efficiently [23]. For instance, the marginalized denoising autoencoder and a nonnegative sparse autoencoder were combined in the form of a cascade method to conduct SU, exhibiting robustness and intrinsic self-adaptation capabilities [24]. Also, several other autoencoder-based SU methods have been developed, including the stacked nonnegative sparse autoencoder [25], untied denoising autoencoder (uDAS) [26], sparse autoencoder network [27], deep autoencoder network [28], convolutional autoencoder [29], and a neural network autoencoder [30].

More recently, the LMM has been expanded to an augmented model to better address spectral variabilities, which can significantly improve the performance on abundance estimation [6]. However, all the aforementioned linear models are based on a rough assumption that may not hold in practice [31]. Nonlinear mixture models (NLMMs), instead, aim at characterizing the interactions between the sunlight scattered by multiple materials, being at an intimate or multilayered level [32], [33].

1) For intimate mixtures, the interactions consist of photons emitted from molecules of one material absorbed by molecules of another material, which may in turn emit more photons [34]–[37]. Generally, the models that assume intimate mixtures are more faithful to the imaging mechanism of the instrument. However, it is worth noting that these models are strongly sensitive to some inherent parameters, which are often determined by the knowledge of the geometric positioning of the sensor. Such dependence upon external parameters brings great difficulties when implementing the inversion [32].
2) For multilayered mixtures, the corresponding models hold that multiple interactions among the scatterers are at different layers, and lead to an infinite sequence of powers of products of reflectance, which are formalized to a combination of the LMM and additional high-order terms [31], [38]. In this case, high-order scattering (derived from the high-order interactions of photons between different materials), is negligibly weak.

NLMM approaches generally estimate the impact of the scattering interactions, such as the generalized bilinear model (GBM) unmixing [39], semi-NMF (Semi-NMF) [40], or robust NMF (rNMF) [41]. The GBM and Semi-NMF are supervised methods which need to know endmembers in advance, while rNMF is a fully unsupervised algorithm. However, the nonlinear interactions on abundances still cannot be obtained by rNMF. Moreover, many kernel-based nonlinear unmixing methods have been developed by using kernel functions and physically inspired models [42], [43], opening up new avenues for nonlinear unmixing [31].

Unfortunately, most NLMM approaches are too complex and difficult to exploit in real scenarios, since they require detailed knowledge about the physical interactions between the sunlight scattered by multiple materials. For simplicity, bilinear mixture models (BMMs) have been proposed and used more commonly in practice; these models consider the second-order scattering of photons between two materials [44]. Despite the fact that they neglect high-order scattering, BMMs can provide very good approximations to the sunlight scattering mechanism in real scenarios [32]. Under the BMM, the formula of SU can be regarded as a combination of linear and bilinear terms, in which the linear components explain endmembers and abundances, and bilinear components reflect the second-order scattering interactions [39].

The development of methods able to model both linear and nonlinear interactions is quite challenging, as they involve different tasks in the optimization procedure [45]. In general, any optimization task is based on a training procedure in which the model is iteratively trained (with a particular metric) until it converges to a stable solution [46]. However, if the model only focuses on one single task, it generally ignores some information which may be helpful for the learning of the other related tasks [47]. To solve the above problem, multitask learning (MTL) was developed to improve the optimization performance by utilizing shared information among different tasks [48], [49]. As a common training strategy, MTL has been successfully applied in many fields, such as speech recognition, computer vision, and natural language processing [48], [50], and also have wide application prospect in the hyperspectral unmixing. Recently, a linear unmixing approach, by using MTL and spectral-spatial information, has been introduced in [51], demonstrating the good potential for hyperspectral unmixing.

In this article, we develop an unsupervised unmixing approach based on the hierarchical BMM, which can estimate the endmember signatures, the abundance fractions, the interaction between abundance fractions, and the interaction outliers caused by bilinear components. Here, the interaction outliers physically represent the scattering interactions between materials in a hyperspectral image. To do so, our method operates within an MTL framework to combine two different tasks. Task 1 estimates endmembers and abundances via a deep autoencoder, while task 2 adopts another deep autoencoder to update the bilinear components, simultaneously. The encoders are able to decompose hyperspectral data and the decoders can reconstruct the data. It should be noted that the two deep autoencoders are trained in a mutually interdependent manner, and the stopping criterion is based on the relative reconstruction error (RE). Besides, we also integrate some aspects of the deep semi-NMF [52] into the aforementioned autoencoders, in order to learn some hidden representations from the data and further improve the performance of the optimization.

Compared with other nonlinear unmixing algorithms, the main advantages of our newly proposed method can be summarized as follows.

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1) We provide a fully unsupervised unmixing approach that does not require any prior knowledge.
2) Its unmixing performance is enhanced by adopting a hierarchical framework, able to learn multiple hidden representations.
3) By taking advantage of the MTL framework, our unmixing (based on the BMM) is split into two tasks which can be executed in parallel.
4) Our method can estimate endmembers and abundances, as well as the interaction abundances and interaction outliers, simultaneously.

The remainder of this article is organized as follows. In Section II, we review some related works. Section III describes the proposed method in detail. In Section IV, synthetic and real hyperspectral images are used for evaluation purposes, allowing us to conduct a quantitative comparison with other state-of-the-art unmixing algorithms. Section V concludes this article with some remarks and hints at plausible future research.

II. Related Works

Our proposed approach employs two deep autoencoders to execute blind source separation on the observed data, which will bring multiple hidden representations associated with the BMM. The linear and bilinear components in the BMM correspond to the two hierarchies, respectively.

A. BMM

Let $\mathbf{Y} \equiv [\mathbf{y}_1, \ldots, \mathbf{y}_n] \in \mathbb{R}^{d\times n}$ be a hyperspectral image with n spectral pixels and d spectral bands. In this work, the constraints of the BMM are derived from the Fan model [53]. For a hyperspectral image, the matrix form of the BMM is denoted as

$$\begin{aligned} &\mathbf{Y} = \mathbf{EA} + \mathbf{DB} + \mathbf{N} \\ &\text{s.t. } \mathbf{A} \geq \mathbf{0}, \mathbf{B} \geq \mathbf{0}, \mathbf{1}_c^\top \mathbf{A} = \mathbf{1}_n^\top \end{aligned} \tag{1}$$

where $\mathbf{E} \equiv [\mathbf{e}_1, \ldots, \mathbf{e}_c] \in \mathbb{R}^{d\times c}$ represents the mixing matrix with c endmembers, $\mathbf{A} \equiv [\mathbf{a}_1, \ldots, \mathbf{a}_n] \in \mathbb{R}^{c\times n}$ denotes abundance matrix. For this product term, the two constraints $\mathbf{A} \geq 0$ and $\mathbf{1}_c^\top \mathbf{A} = \mathbf{1}_n^\top$ are commonly known as the abundance nonnegativity constraint (ANC) and abundance sum-to-one constraint (ASC) with $\mathbf{1}_c = [1, 1, \ldots, 1]^\top \in \mathbb{R}^c$, which stem from a physical interpretation of the abundance fractions [54]. The matrix $\mathbf{D} \equiv [\mathbf{d}_1, \ldots, \mathbf{d}_l] \in \mathbb{R}^{d\times l}$ contains l virtual endmembers, and $\mathbf{B} \equiv [\mathbf{b}_1, \ldots, \mathbf{b}_l] \in \mathbb{R}^{l\times n}$ denotes an interaction abundance matrix, with $l = c(c-1)/2$. The matrix $\mathbf{N} \in \mathbb{R}^{d\times n}$ denotes an error matrix that may affect the imaging process (e.g., noise). Note that the notation $[\cdot]^\top$ stands for the transposition of a matrix or vector. Following [40], we assume that a virtual endmember $\mathbf{d}_{(i,j)}$ is initialized by the corresponding endmembers. With these assumptions in mind, we have

$$\mathbf{d}_{(i,j)} = \mathbf{e}_i \odot \mathbf{e}_j \tag{2}$$

where $\odot$ is the Hadamard product, $\mathbf{e}_i$ is the ith endmember signature in $\mathbf{E}$, $\forall\, i \in \{1, \ldots, c-1\}$, $j \in \{i+1, \ldots, c\}$. Let $b_{(i,j),q}$ be an interaction abundance, initialized by

$$b_{(i,j),q} = a_{i,q} a_{j,q} \tag{3}$$

where $b_{(i,j),q}$ is an element of $\mathbf{b}_q$ at the qth column of $\mathbf{B}$, and $a_{i,q}$ and $a_{j,q}$ are the ith element and the jth element of the qth column of $\mathbf{A}$, respectively.

B. Hierarchical BMM

In (1), let $\mathbf{Y}_L \in \mathbb{R}^{d\times n}$ be the linear term with

$$\mathbf{Y}_L = \mathbf{EA}$$

and let $\mathbf{Y}_N \in \mathbb{R}^{d\times n}$ be the bilinear term with

$$\mathbf{Y}_N = \mathbf{DB}.$$

Following [52], [55], and [56], we have

$$\begin{aligned} \mathbf{Y}_L &= \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_m \mathbf{A}_m \\ \mathbf{Y}_N &= \mathbf{D}_1 \mathbf{D}_2 \cdots \mathbf{D}_m \mathbf{B}_m \end{aligned} \tag{4}$$

respectively, where m is the number of hidden layers of the encoder or decoder. For the autoencoders used, the structure of the encoder is the same as that of the decoder. In the hierarchy, $\mathbf{E}_1, \ldots, \mathbf{E}_m$ denote weight matrices between adjacent layers, and $\mathbf{A}_1, \ldots, \mathbf{A}_m$ represent hidden layers. Similarly, $\mathbf{D}_1, \ldots, \mathbf{D}_m$ are weight matrices, and $\mathbf{B}_1, \ldots, \mathbf{B}_m$ represent hidden layers. Note that the sizes of these matrices may not be the same, $\mathbf{E}_1 \in \mathbb{R}^{d\times c}$, $\mathbf{D}_1 \in \mathbb{R}^{d\times l}$, $\mathbf{E}_k \in \mathbb{R}^{c\times c}$, $\mathbf{D}_k \in \mathbb{R}^{l\times l}$, with $k = 2, \ldots, m$. Following [52], the hierarchies for the multiple hidden representations are given as:

$$\begin{aligned} \mathbf{A}_{m-1} &\approx \mathbf{E}_m \mathbf{A}_m \\ &\vdots \\ \mathbf{A}_2 &\approx \mathbf{E}_3 \cdots \mathbf{E}_m \mathbf{A}_m \\ \mathbf{A}_1 &\approx \mathbf{E}_2 \cdots \mathbf{E}_m \mathbf{A}_m \end{aligned} \tag{5}$$

and

$$\begin{aligned} \mathbf{B}_{m-1} &\approx \mathbf{D}_m \mathbf{B}_m \\ &\vdots \\ \mathbf{B}_2 &\approx \mathbf{D}_3 \cdots \mathbf{D}_m \mathbf{B}_m \\ \mathbf{B}_1 &\approx \mathbf{D}_2 \cdots \mathbf{D}_m \mathbf{B}_m. \end{aligned} \tag{6}$$

For the linear term, $\mathbf{Y}_L$ is first decomposed into $\mathbf{E}_1$ and $\mathbf{A}_1$, $\mathbf{Y}_L \approx \mathbf{E}_1 \mathbf{A}_1$, and the hidden layers are then decomposed one-by-one, $\mathbf{A}_q \approx \mathbf{E}_{q+1} \mathbf{A}_{q+1}$, $\mathbf{A}_q \in \mathbb{R}^{c\times n}$, $q = 2, \ldots, m-1$. In the same way, the bilinear term is expressed as $\mathbf{Y}_N \approx \mathbf{D}_1 \mathbf{B}_1$, $\mathbf{B}_q \approx \mathbf{D}_{q+1} \mathbf{B}_{q+1}$, $\mathbf{B}_q \in \mathbb{R}^{l\times n}$. For the linear components, $\mathbf{E}$ and $\mathbf{A}$ are given by the following mathematical formulations:

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_1 \cdots \mathbf{E}_{m-1} \mathbf{E}_m \\ \mathbf{A} &= \mathbf{A}_m. \end{aligned} \tag{7}$$

In the same way as the bilinear components, $\mathbf{D}$ and $\mathbf{B}$ are given as follows:

$$\begin{aligned} \mathbf{D} &= \mathbf{D}_1 \cdots \mathbf{D}_{m-1} \mathbf{D}_m \\ \mathbf{B} &= \mathbf{B}_m. \end{aligned} \tag{8}$$

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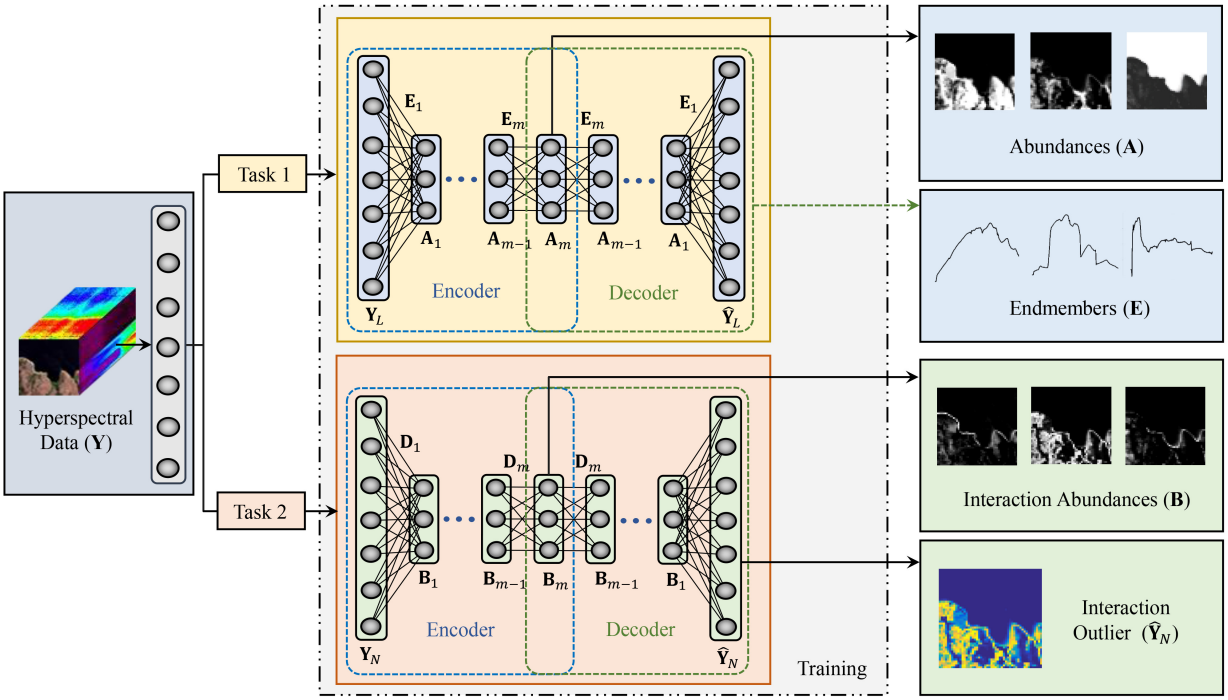


Fig. 1. Flowchart of the proposed method. Task 1 is used for updating the linear components, estimating the endmember signatures $\mathbf{E}$ and abundance fractions $\mathbf{A}$. Task 2 responds to the bilinear components, obtaining the interaction abundances $\mathbf{B}$ and the interaction outliers $\widehat{\mathbf{Y}}_N$.

III. Proposed Method

In the proposed method, the autoencoders are trained together by minimizing the discrepancy between the original data and the reconstructed ones. The training of our method contains two stages: pretraining and fine-tuning. The two autoencoders first pretrain all factors layer-by-layer, respectively, and then all factors are fine-tuned. In the training, each node of a hidden layer in task 2 is activated by the Hadamard product between any two nodes of the corresponding hidden layer in task 1. Let $\widehat{\mathbf{Y}} = \mathbf{EA} + \mathbf{DB}$ be the reconstruction of the image $\mathbf{Y}$, $\widehat{\mathbf{Y}} \equiv [\widehat{\mathbf{y}}_1, \ldots, \widehat{\mathbf{y}}_n] \in \mathbb{R}^{d\times n}$. The fine-tuning would stop when the RE reaches convergence, and the RE is denoted as

$$\mathrm{RE}\left(\{\mathbf{y}_g\}_{g=1}^n, \{\widehat{\mathbf{y}}_g\}_{g=1}^n\right) = \frac{1}{n}\sum_{g=1}^{n}\sqrt{\|\mathbf{y}_g - \widehat{\mathbf{y}}_g\|_2^2}. \tag{9}$$

In the MTL framework, the two autoencoders correspond to two tasks, and each autoencoder has $2m$-1 hidden layers, because the encoder and decoder have the same structure, as illustrated in Fig. 1. In task 1, the endmembers are obtained from weight matrices of the decoder, and the abundance matrix corresponds to $\mathbf{A}_m$ in the middle hidden layer. While in task 2, interaction outliers are estimated from the last layer of the decoder, and the interaction abundances correspond to $\mathbf{B}_m$ in the middle hidden layer. For the autoencoder of the corresponding linear term, the number of nodes in each hidden layer is set to the number of endmembers. For the autoencoder of the corresponding bilinear term, the number of nodes in each hidden layer amounts to the number of virtual endmembers. The number of input and output nodes of the two autoencoders is equal to the number of pixels in the scene.

A. Task 1 for Estimating Linear Components

Task 1 is used for estimating the endmember signatures and the abundance fractions, and the objective function of the subproblem is written as

$$\mathcal{C}_{\text{Task1}} = \frac{1}{2}\left(\|\mathbf{Y} - \mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_m\mathbf{A}_m - \mathbf{DB}\|_F^2 + \mu\Re(\mathbf{A}_k)\right) \tag{10}$$

where $\|\cdot\|_F^2$ denotes the Frobenius norm, μ is the regularization coefficient of task 1. For this task, $\mathbf{D}$ and $\mathbf{B}$ are regarded as the constant matrices. With respect to $\mathbf{A}_k$, each of its rows corresponds to a nodes of the kth hidden layer. Note that, although $\mathbf{A}_k$ is related to abundances, it is not directly equivalent to abundance fractions. Following [52] and [57], $\Re(\mathbf{A}_k) = \mathrm{Tr}(\mathbf{A}_k\mathbf{L}\mathbf{A}_k^\top)$ controls the smoothness of $\mathbf{A}_k$, where $\mathrm{Tr}(\cdot)$ denotes the trace norm. Here, $\mathbf{L} = \mathbf{X} - \mathbf{W}$ is the graph Laplacian matrix obtained from the radial basis kernel function. $\mathbf{W}$ is called graph connection matrix, in which each element w_{ph} represents the graph connection between pth pixel and hth pixel, and satisfies $w_{ph} = w_{hp}$. $\mathbf{X}$ is a diagonal matrix, in which each diagonal element is the column sum of $\mathbf{W}$. In this autoencoder, all hidden layers share $\mathbf{L}$ and the nodes associated with the abundances. Following [58], the graph connection is denoted as:

$$w_{ph} = \begin{cases} \exp\left(-\dfrac{\|\mathbf{y}_p - \mathbf{y}_h\|^2}{2\sigma^2}\right), & \text{if } p \neq h \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where $\mathbf{y}_p$ and $\mathbf{y}_h$ are vectors of $\mathbf{Y}$, $\forall\ p \in \{1, \ldots, n-1\}$, $h \in \{p+1, \ldots, n\}$. The parameter σ can be estimated from a conjugate inverse Gamma distribution [59]. The generation of

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$\mathbf{W}$ is defined as follows:

$$\mathbf{W} = \begin{bmatrix} 0 & w_{12} & \cdots & w_{1n} \\ w_{21} & 0 & \cdots & w_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n1} & w_{n2} & \cdots & 0 \end{bmatrix}. \tag{12}$$

Summing up w_{ph} in each row of $\mathbf{W}$, the diagonal matrix $\mathbf{X}$ is obtained as

$$\mathbf{X} = \begin{bmatrix} w_{12} + \cdots w_{1n} & 0 & \cdots & 0 \\ 0 & w_{21} + \cdots w_{2n} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_{n1} + \cdots w_{n(n-1)} \end{bmatrix}. \tag{13}$$

In order to meet the ASC and accelerate convergence, we use the multiplicative update rule (MUR) to obtain $\mathbf{E}_1$ and $\mathbf{A}_1$, where the adopted MUR is formula from [60]

$$\begin{aligned} \mathbf{E}_1 &\leftarrow \mathbf{E}_1 \odot \left(\mathbf{Y}\mathbf{A}_1^\top\right) \cdot / \left(\mathbf{E}_1\mathbf{A}_1\mathbf{A}_1^\top\right) \\ \mathbf{A}_1 &\leftarrow \mathbf{A}_1 \odot \left(\widetilde{\mathbf{A}}_1^\top\widetilde{\mathbf{Y}}\right) \cdot / \left(\widetilde{\mathbf{E}}^\top\widetilde{\mathbf{E}}\mathbf{A}_1\right) \end{aligned} \tag{14}$$

where $\widetilde{\mathbf{Y}}$ and $\widetilde{\mathbf{A}}_1$ are the augmented matrices for satisfying the ASC constraint, and $\cdot/$ represents an elementwise division. As a result, we have

$$\widetilde{\mathbf{Y}} = \begin{bmatrix} \mathbf{Y} \\ \delta\mathbf{1} \end{bmatrix}, \quad \widetilde{\mathbf{A}}_1 = \begin{bmatrix} \mathbf{A}_1 \\ \delta\mathbf{1} \end{bmatrix} \tag{15}$$

where δ is able to control the impact of the ASC constraint. For the deep autoencoder, δ can affect $\mathbf{A}_1$, and then it can also affect $\mathbf{A}_k$, because $\mathbf{A}_1$ is associated with $\mathbf{A}_k$. After evaluating the performance of our algorithm with different values of δ, we empirically set $\delta = 23$ in this work.

With respect to other weights and hidden layers, gradient descent (GD) is employed to implement the optimization. The first-order partial derivatives of $\mathcal{C}_{\text{Task1}}$ with respect to $\mathbf{A}_k$ are defined as

$$\nabla_{\mathbf{A}_k}\mathcal{C}_{\text{Task 1}} = \mathbf{E}_k^\top\boldsymbol{\Psi}^\top(\boldsymbol{\Psi}\mathbf{E}_k\mathbf{A}_k - \mathbf{Y}) + \mu\mathbf{A}_k\mathbf{L} \tag{16}$$

where $\boldsymbol{\Psi}$ is defined as $\boldsymbol{\Psi} = \mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_{k-1}$. Similarly, we have the partial derivative of $\mathcal{C}_{\text{Task1}}$ for $\mathbf{E}_k$

$$\nabla_{\mathbf{E}_k}\mathcal{C}_{\text{Task 1}} = \boldsymbol{\Psi}^\top(\boldsymbol{\Psi}\mathbf{E}_k\widetilde{\mathbf{A}}_k - \mathbf{Y})\widetilde{\mathbf{A}}_k^\top \tag{17}$$

where $\widetilde{\mathbf{A}}_k$ is a reconstruction, $\widetilde{\mathbf{A}}_k = \mathbf{E}_{k+1}\ldots\mathbf{E}_m\mathbf{A}_m$. The derivations of (16) and (17) are shown in the Appendix.

Afterward, $\mathbf{E}_k$ and $\mathbf{A}_k$ are updated via the GD. Considering the ANC, we add the rectified linear unit ReLU activation to the kth hidden layer

$$\begin{aligned} \mathbf{A}_k &\leftarrow \text{ReLU}\left(\mathbf{A}_k - \eta_A\nabla_{\mathbf{A}_k}\mathcal{C}_{\text{Task1}}\right) \\ \mathbf{E}_k &\leftarrow \mathbf{E}_k - \eta_E\nabla_{\mathbf{E}_k}\mathcal{C}_{\text{Task1}} \end{aligned} \tag{18}$$

where $\text{ReLU}(\cdot) = \max(\cdot, 0)$, the corresponding learning rates η_E and η_A are estimated by Armijo rule [61]. Finally, we obtain the abundances $\mathbf{A}$ and endmembers $\mathbf{E}$ as follows:

$$\begin{aligned} \mathbf{A} &= \mathbf{A}_m \\ \mathbf{E} &= \mathbf{E}_1\cdots\mathbf{E}_{m-1}\mathbf{E}_m. \end{aligned} \tag{19}$$

B. Task 2 for Estimating Bilinear Components

Task 2 estimates the bilinear components, i.e., the interactions between abundance fractions and outlier. In hyperspectral images, the scattering interactions are mainly located in the boundaries and mixing regions between materials [41]. Therefore, compared with the material distributions, the scattering interactions usually have low-rank property. The objective function of the subproblem is defined as

$$\mathcal{C}_{\text{Task 2}} = \frac{1}{2}\left(\|\mathbf{Y} - \mathbf{E}\mathbf{A} - \mathbf{D}_1\mathbf{D}_2\cdots\mathbf{D}_m\mathbf{B}_m\|_F^2 + \upsilon\|\mathbf{B}_m\|_*\right) \tag{20}$$

where $\|\mathbf{B}_m\|_* = \text{Tr}(\mathbf{B}_m{}^\top\mathbf{B}_m)^{1/2}$ is used to impose low-rank, $\|\cdot\|_*$ denotes the nuclear norm, and υ is the regularization coefficient. Following the update rule of the deep semi-NMF method in [52], we adopt it to update $\mathbf{B}_r \in \mathbb{R}^{l\times n}$:

$$\mathbf{B}_r \leftarrow \mathbf{B}_r \odot \sqrt{\frac{\left[\boldsymbol{\Phi}^\top(\mathbf{Y} - \mathbf{E}\mathbf{A})\right]^+ + \left[\boldsymbol{\Phi}^\top\boldsymbol{\Phi}\right]^-\mathbf{B}_r}{\left[\boldsymbol{\Phi}^\top(\mathbf{Y} - \mathbf{E}\mathbf{A})\right]^- + \left[\boldsymbol{\Phi}^\top\boldsymbol{\Phi}\right]^+\mathbf{B}_r}} \tag{21}$$

where $\boldsymbol{\Phi} = \mathbf{D}_1\mathbf{D}_2\cdots\mathbf{D}_{r-1}$, $r = 1, \ldots, m-1$. Following [62], $[\cdot]^+$ and $[\cdot]^-$ are the positive and negative parts of the matrix, respectively. The update rule of $\mathbf{D}_r \in \mathbb{R}^{l\times l}$ is denoted as

$$\mathbf{D}_r \leftarrow \mathbf{D}_r \odot \left(\boldsymbol{\Phi}^\top(\mathbf{Y} - \mathbf{E}\mathbf{A})\mathbf{B}_r^\top\right) ./ \left(\boldsymbol{\Phi}^\top\boldsymbol{\Phi}\mathbf{D}_r\mathbf{B}_r\mathbf{B}_r^\top\right). \tag{22}$$

In task 2, the mth hidden layer needs a constraint to ensure a low-rank and nonnegative attribute on interaction abundances, thus we first set $\mathbf{D}_m = \mathbf{D}_{m-1}$ and $\mathbf{B}_m = \mathbf{B}_{m-1}$ to provide the initial values for $\mathbf{D}_m$ and $\mathbf{B}_m$. According to the constraint in [63], we adopt the following rule for $\mathbf{B}_m$:

$$\mathbf{B}_m \leftarrow \text{sgn}(\widetilde{\mathbf{B}}_m) \odot \text{ReLU}\left(\widetilde{\mathbf{B}}_m - \boldsymbol{\Lambda}_{\widetilde{B}}\right) \tag{23}$$

where $\text{sgn}(\cdot)$ represents the sign function, and the threshold is given as $\Lambda_{\widetilde{B}} = \lambda./(|\widetilde{\mathbf{B}}_m| + \lambda)$ with $\lambda = 0.001$. The hyperparameter λ avoids that the denominator is a zero matrix

$$\widetilde{\mathbf{B}}_m = \mathbf{U}\ \text{diag}(\widetilde{\mathbf{S}})\ \mathbf{V}^T \tag{24}$$

where $\mathbf{U}, \mathbf{S}, \mathbf{V}$ are obtained by a singular value decomposition (SVD) of $\mathbf{B}_m$, i.e., $[\mathbf{U}, \mathbf{S}, \mathbf{V}] = \text{SVD}(\mathbf{B}_m)$ and $\widetilde{\mathbf{S}} = \max(|\text{diag}(\mathbf{S})| - \Lambda_S, 0)$, with $\Lambda_S = \lambda./(|\text{diag}(\mathbf{S})| + \lambda)$. Let $\mathbf{D} = \mathbf{D}_1\cdots\mathbf{D}_{m-1}\mathbf{D}_m$. Then, we obtain the interaction abundances and interaction outliers as follows:

$$\begin{aligned} \mathbf{B} &= \mathbf{B}_m \\ \widehat{\mathbf{Y}}_N &= \mathbf{D}\mathbf{B}. \end{aligned} \tag{25}$$

C. Implementation Details

In this section, we provide the implementation details for our algorithm. First and foremost, we emphasize that we assume the number of endmembers to be known in advance. In addition, the weights and hidden layers are pretrained using the strategy in [20]. In task 1, $\mathbf{E}_1$ is obtained by running a pure pixel-based unmixing algorithm (e.g., VCA). Meanwhile, $\mathbf{A}_1$ is obtained by using the fully constrained least square (FCLS) method in [54]. Afterward, $\mathbf{E}_k$ and $\mathbf{A}_k$ are initialized by using the same strategy. With respect to task 2, the weights and hidden layers are initialized by using the corresponding factors resulting from task 1. More specifically, the initialization of

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$\mathbf{d}_{(i,j),1}$ is calculated by $\mathbf{d}_{(i,j),1} = \mathbf{e}_{i,1} \odot \mathbf{e}_{j,1}$, where $\mathbf{e}_{i,1}$ and $\mathbf{e}_{j,1}$ are two column vectors in $\mathbf{E}_1$, $\forall\ i \in \{1, \ldots, c-1\}$, $j \in \{i+1, \ldots, c\}$. Then, $\mathbf{D}_1$ is composed by $\mathbf{d}_{(i,j),1}$. Here, we assume that $b_{(i,j),q,1}$ is an element of the qth column of $\mathbf{B}_1$, $b_{(i,j),q,1} = a_{i,q,1}a_{j,q,1}$, where $a_{i,q,1}$ and $a_{j,q,1}$ are the ith element and the jth element of the qth column of $\mathbf{A}_1$, respectively. Afterward, $\mathbf{B}_1$ is composed by $b_{(i,j),q,1}$, The weights and nodes of other layers are then initialized using this same strategy. Finally, the fine-tuning of each layer is performed until the RE in (9) achieves convergence. A pseudocode of the proposed method is illustrated in Algorithm 1. In lines 1 and 2, the pretraining is implemented via VCA and FCLS. In lines 3 and 4, the weights and nodes of hidden layers in task 1 are updated by fine-tuning. Lines 5–7 carry out the update of weights and nodes in task 2 (by means of fine-tuning). Line 8 obtains the endmember and abundance matrices. Line 9 generates the interaction abundance and outlier matrices.

Algorithm 1 Pseudocode of the Proposed Approach

Input: Hyperspectral data $\mathbf{Y}$, Layer number m.
Output: Endmembers $\mathbf{E}$, Abundances $\mathbf{A}$, Interaction abundances $\mathbf{B}$, Interaction outliers $\widehat{\mathbf{Y}}_N$.
for all layers **do**
/* *Initialization* */
1. Run VCA and FCLS for the initialization of task 1.
2. Initialize task 2 by using the factors resulting from task 1.
end for
repeat
/* *Task 1 (for linear components)* */
for all layers **do**
3. Update $\mathbf{E}_1$ and $\mathbf{A}_1$ in Eq. (14).
while reach the threshold of Armijo rule **do**
4. Update $\mathbf{E}_k$ and $\mathbf{A}_k$ in Eq. (18).
end while
end for
/* *Task 2 (for bilinear components)* */
for all layers **do**
5. Update $\mathbf{B}_r$ in Eq. (21).
6. Update $\mathbf{D}_r$ in Eq. (22).
7. Update $\mathbf{B}_m$ in Eq. (23).
end for
8. Obtain $\mathbf{E}$ and $\mathbf{A}$ in Eq. (19).
9. Obtain $\mathbf{B}$ and $\widehat{\mathbf{Y}}_N$ in Eq. (25).
until RE reaches convergence

IV. Experimental Results and Analysis

The effectiveness of the proposed method called hereinafter deep multitask bilinear unmixing (DMBU) is evaluated by using both synthetic and real hyperspectral data sets. In this article, four metrics are adopted for the assessment of results, including the spectral angle distance (SAD), root mean square error (RMSE), RE, and variance inflation factor (VIF).

The SAD is used to measure the quality of the estimated endmember, which is defined as

$$\mathrm{SAD}(\mathbf{e}, \widehat{\mathbf{e}}) = \arccos\left(\frac{[\mathbf{e}, \widehat{\mathbf{e}}]}{\|\mathbf{e}\| \cdot \|\widehat{\mathbf{e}}\|}\right) \tag{26}$$

where $\widehat{\mathbf{e}}$ and $\mathbf{e}$ denote the estimated endmember and library spectrum, respectively, and the SAD is specified in radians.

The quality of the estimated abundances is measured by using the RMSE defined as

$$\mathrm{RMSE}(\widehat{\mathbf{a}}_g,\ \mathbf{a}_g) = \frac{1}{n}\sum_{g=1}^{n}\sqrt{\|\ \mathbf{a}_g - \widehat{\mathbf{a}}_g\|_2^2} \tag{27}$$

where $\widehat{\mathbf{a}}_g$ and $\mathbf{a}_g$ are the corresponding estimated and actual abundance fractions. The RE was given in (9). Note that the RE should not be directly regarded as a metric for measuring the accuracy of unmixing results. Instead, it is used as a complementary evaluation, following [40] and [64].

Moreover, we use the VIF to measure the degree of collinearity, according to the relevant works in [64]–[66]. This metric is defined as

$$\mathrm{VIF}_i = \frac{1}{1 - X_i^2} \tag{28}$$

where X_i^2 denotes the multiple correlation between the ith variable (an endmember, or a virtual endmember) and other explanatory variables. A detailed explanation of the VIF is available in [65]. Note that, in this article, the VIF denotes the mean of all VIF_i.

The remainder of this section is organized as follows. Section IV-A describes the experiments with synthetic data sets. These experiments focus on the network parameters, the collinearity impact, and a comparative algorithm assessment. In Section IV-B, some real hyperspectral images are used to further evaluate the effectiveness of our newly proposed DMBU. Note that all experiments have been performed in a workstation with Intel Core I7 CPU and 16 GB of RAM.

A. Experiments With Synthetic Data

The synthetic data set is generated using a BMM. Three pure spectral signatures are randomly selected from the United States Geological Survey (USGS) spectral library.[1] Each pixel in the synthetic data set has 224 spectral bands, covering the spectral range from 0.4 to 2.5 μm. The abundance settings follow a random distribution, and the maximum abundance purity of the synthetic data is set to 0.9, i.e., all pure pixels are removed. In our synthetic data experiments, we simulated several different scenarios: different numbers of pixels and different signal-noise ratio (SNR) levels. An illustrative scatterplot is given in Fig. 2, where the original data are projected onto the first three principal components (PCs) to facilitate visualization.

1) Parameter Analysis: In this test, the performance of network parameters is verified under different settings. The experimental results with respect to the coefficients in (10) and (20) are displayed in Table I. For the data used in Table I, the number of pixels is 3364, and the SNR is 40 dB. For the autoencoders, the number of hidden layers is set to $m = 3$. The obtained results indicate that the best performance is obtained within a relatively wide range of values, i.e., $0.1 < \mu < 0.9$, $0.05 < \upsilon < 0.9$. Moreover, these results indicate that the

[1] https://speclab.cr.usgs.gov/spectrallib.html

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TABLE I

MEAN SADS (IN RADIANS), RMSES AND RES OBTAINED BY THE PROPOSED DMBU BY USING DIFFERENT BALANCE COEFFICIENTS, FOR A SYNTHETIC DATA SET WITH 3364 PIXELS AND SNR OF 40 DB

		SAD	RMSE	RE
$\mu = 0.9$	$\upsilon = 0.9$	0.0161	0.4208	0.0148
	$\upsilon = 0.5$	0.0149	0.4004	0.0127
	$\upsilon = 0.1$	0.0127	0.3813	0.0103
	$\upsilon = 0.05$	0.0130	0.3906	0.0105
$\mu = 0.5$	$\upsilon = 0.9$	0.0154	0.4255	0.0119
	$\upsilon = 0.5$	0.0144	0.4192	0.0114
	$\upsilon = 0.1$	0.0124	0.3702	0.0091
	$\upsilon = 0.05$	0.0129	0.3916	0.0092
$\mu = 0.1$	$\upsilon = 0.9$	0.0168	0.4569	0.0175
	$\upsilon = 0.5$	0.0135	0.4371	0.0087
	$\upsilon = 0.1$	0.0149	0.4424	0.0092
	$\upsilon = 0.05$	0.0184	0.4406	0.0118
$\mu = 0.05$	$\upsilon = 0.9$	0.0134	0.4221	0.0094
	$\upsilon = 0.5$	0.0141	0.4407	0.0101
	$\upsilon = 0.1$	0.0137	0.4445	0.0090
	$\upsilon = 0.05$	0.0152	0.4373	0.0086

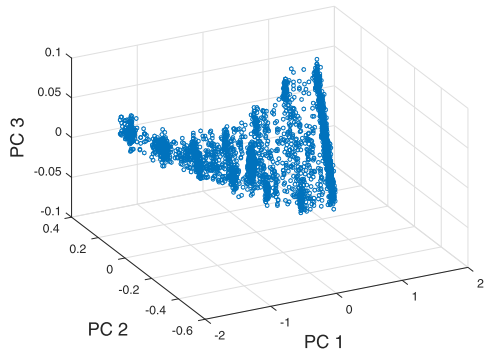


Fig. 2. Scatterplot of one of our simulated data sets, with 3364 pixels and SNR of 40 dB.

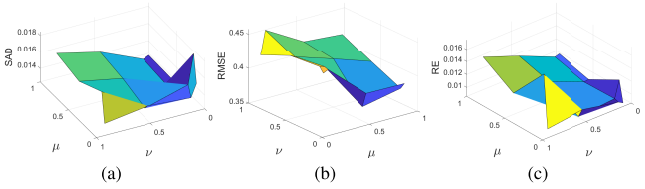


Fig. 3. Results of the proposed DMBU with respect to balance coefficients μ and υ. (a) SAD. (b) RMSE. (c) RE.

proposed approach is quite insensitive in the aforementioned range. Fig. 3 visualizes the quantitative results obtained with respect to two balance coefficients. In the subsequent experiments, the coefficients are set as $\mu = 0.5$ and $\upsilon = 0.1$.

For autoencoder, the number of layers is also an important factor defining unmixing performance. Table II shows the differences in the obtained results when we assign a different number of layers in the proposed approach. In this test, the number of pixels is set to 3364, with an SNR of 40 dB. From Table II, we can observe that the results are better when m is set to 3 or 4. However, the performances are degraded

TABLE II

MEAN SADS (IN RADIANS), RMSES, RES AND COMPUTATION TIME (IN SECONDS) OBTAINED BY THE PROPOSED DMBU WITH DIFFERENT m

	m=1	m=2	m=3	m=4	m=5	m=6
SAD	0.0118	0.0114	0.0109	0.0107	0.0112	0.0110
RMSE	0.4030	0.3107	0.2662	0.2649	0.2719	0.2646
RE	0.0145	0.0135	0.0129	0.0128	0.0131	0.0127
Time	21.6308	34.9588	62.8081	85.2175	104.0968	137.3905

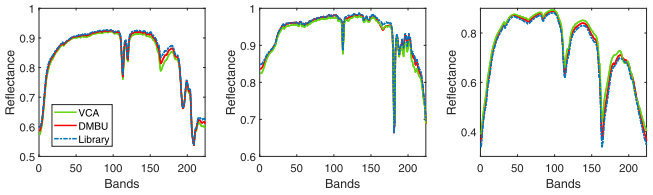


Fig. 4. Comparison between the estimated endmembers, along with the corresponding library signatures.

when the parameter m is more than 4. In addition, more hidden layers mean more computation time to implement the proposed approach. Considering the performance results and computation time in Table II, we set $m = 3$ in the subsequent experiments.

In order to graphically illustrate the effectiveness of the proposed method, the estimated endmembers via DMBU are shown in Fig. 4. The results obtained by the VCA are also displayed in Fig. 4, considering that they are used to initialize the proposed approach. In this test, the data include again 3364 pixels and an SNR of 40 dB. Fig. 4 shows a graphical comparison between the reference library signatures and the estimated endmembers. Fig. 4 reveals that the endmembers estimated by DMBU are quite similar to the corresponding ones in the library.

Moreover, the proposed DMBU can also estimate abundance fractions. In this test, the number of pixels in the synthetic data is set to 100. The estimated abundances are illustrated in Fig. 5(a). As can be seen in Fig. 5(a), the estimated abundances exhibit a very good match with regards to the ground-truth fractions. Fig. 5(b) shows the original data $\mathbf{Y}$ and the reconstructed data $\widehat{\mathbf{Y}}$. The results in Fig. 5(b) also indicate that the proposed DMBU can lead to very good reconstruction results.

2) Impact of Collinearity Analysis: In multivariate regression, collinearity is a common phenomenon whenever two or more of the variables in a model are moderately or highly correlated [67]. To our knowledge, SU can be regarded as a special optimization problem, so it is hard to absolutely avoid the influence of the collinearity among variables. For SU, the collinearity means that there are correlations among the endmembers, and the adverse impact of the correlations may make regression sensitive to noise and cause a degradation of unmixing performance [64]–[66]. Therefore, it is necessary to evaluate the collinearity effect on virtual endmembers. In this test, we use the VIF to measure the degree of collinearity, under several different scenarios. Table III illustrates the VIFs obtained by the proposed DMBU. The results in Table III indicate that the VIFs of endmembers are

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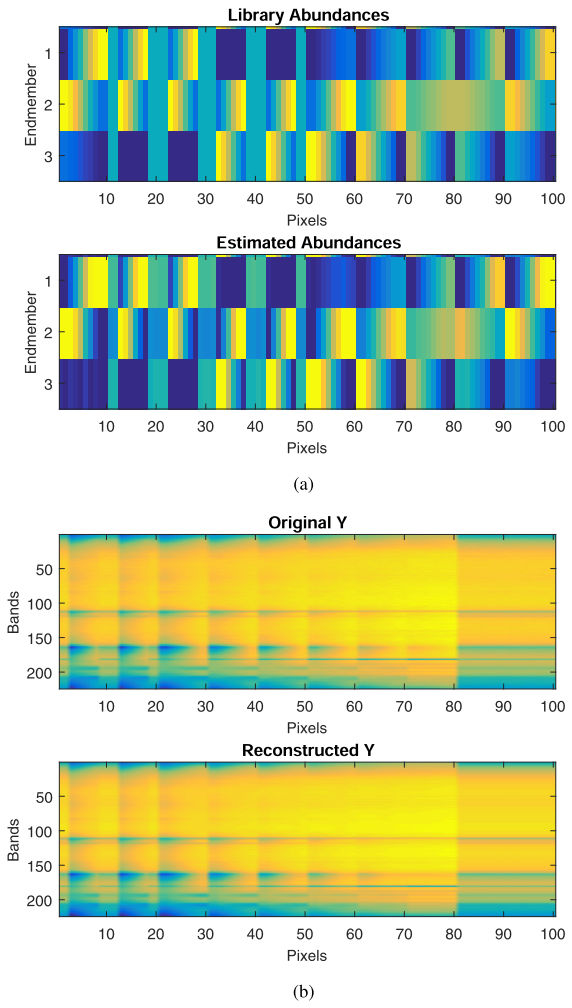


Fig. 5. (a) (Top) Ground-truth abundance. (Bottom) Abundances estimated by the DMBU. (b) (Top) Original data. (Bottom) Reconstructed data by DMBU. Bright yellow: large value. Dark blue: small value.

negligibly small, while the virtual endmembers exhibit high collinearity. The reason is that the collinearity among independent variables is generally not significant. Geometrically speaking, a p-simplex is assumed from a p-dimensional polytope which is the convex hull of its $p+1$ vertices, and these vertices are mutually independent [68], [69]. For a hyperspectral image, the endmembers are located in the vertices of the simplex under the LMM, so the impact of collinearity is generally not significant. However, the virtual endmembers themselves are correlated, because they are produced by the Hadamard product between endmembers. Moreover, Table III also reveals that the collinearity effect decreases with the increase of the noise level.

3) Comparison With Other Algorithms: In this section, our proposed DMBU is evaluated by comparing it with several widely used unmixing algorithms, such as N-FINDR [12], VCA [13], FCLS [54], $L_{1/2}$-NMF [19], uDAS [25], GBM [39], Semi-NMF [40], and rNMF [41]. These unmixing algorithms were chosen as a comparative basis for the following reasons. In the selected methods, N-FINDR, VCA, and FCLS are generally used for the initialization of other methods. $L_{1/2}$-NMF also adopts the MUR to update variables. uDAS is an autoencoder-based method, and it also adopts an empirical coefficient to meet the ASC. N-FINDR and VCA are used for extracting endmember signatures, while FCLS, GBM, and Semi-NMF can only be used for obtaining abundance fractions. In the aforementioned approaches, $L_{1/2}$-NMF, rNMF, uDAS, and DMBU are unsupervised algorithms which are able to estimate endmember signatures and abundance fractions, simultaneously. VCA, N-FINDR, FCLS, uDAS, and $L_{1/2}$-NMF are approaches for linear unmixing. On the contrary, GBM, Semi-NMF, rNMF, and DMBU are nonlinear unmixing approaches. According to the overviews in [8] and [32], the proposed DMBU is a nonlinear unmixing approach because it is based on a BMM.

TABLE III
VIFs OBTAINED BY DMBU IN SEVERAL DIFFERENT SCENARIOS

VIF	100 Pixels		3364 Pixels	
	True Endmembers	Virtual Endmembers	True Endmembers	Virtual Endmembers
SNR=80dB	8.6352	92.2544	9.2458	97.1803
SNR=40dB	8.9893	109.0075	10.9495	89.6224
SNR=30dB	11.6849	193.0777	12.6841	204.4147
SNR=20dB	13.2706	422.2511	14.1632	551.9053

In this comparison experiment, VCA is used to extract endmember signatures before running FCLS, GBM, and Semi-NMF. Note that, for linear unmixing algorithms, the image reconstruction only relates to endmember signatures and abundance fractions. The quantitative results are shown in Table IV, and the results are obtained by the averaging of the results obtained after ten independent Monte Carlo runs. From the results reported in Table IV, it can be observed that the proposed DMBU obtains competitive results when compared with other methods. More specifically, the proposed approach is able to obtain good results on the estimation of endmembers and abundances. This is demonstrated by the fact that it achieves better results in terms of SAD, RMSE, and RE. It is noticeable that the advantages of our approach are suitable for the scenarios in which the SNR level is more than 20 dB. The proposed approach can efficiently address the problem of mixed pixels in most real hyperspectral image scenes, as long as the SNR levels of these images are higher than 20 dB.

B. Experiments With Real Data

In this subsection, the proposed DMBU is applied to four real hyperspectral data sets: Moffett Field [70], Jasper Ridge [71], Urban [72], and Henry Island [73]. The first three data sets have been widely used for testing the performance of unmixing algorithms. The Henry Island data set is also used to conduct subpixel analysis for ecosystems research on mangrove forests. In these experiments, the parameters of DMBU follow the same settings established in the synthetic data experiments, i.e., $\mu = 0.5$, $\upsilon = 0.1$, and $m = 3$.

1) Experiment With Moffett Field Data Set: The Moffett Field data is a real hyperspectral image acquired over Moffett Field, CA, USA, in 1997. It was gathered by the Airborne Visible Infra-Red Imaging Spectrometer (AVIRIS) [70], [74]. In this experiment, the selected subscene comprises

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TABLE IV

MEAN SADS (IN RADIANS), RES AND RMSES, ALONG WITH THE STANDARD DEVIATIONS OBTAINED FROM TEN INDEPENDENT MONTE CARLO RUNS FOR THE CONSIDERED SYNTHETIC DATA. THE BEST RESULTS ARE DISPLAYED IN BOLD

		3364 Pixels							
		N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU
SNR=30 dB	SAD	0.0141±0.02%	0.0099±0.05%	0.0302±6.19%	0.0098±0.14%	0.0099±0.05%	0.0099±0.05%	**0.0091**±0.12%	0.0096±0.08%
	RMSE	0.3894±0.87%	0.3726±0.28%	0.9487±2.52%	0.3705±0.31%	0.3790±0.90%	0.3712±0.35%	0.3670±0.26%	**0.3651**±0.51%
	RE	0.0137±0.01%	0.0141±0.01%	0.0215±0.34%	0.0136±0.09%	0.0132±0.04%	0.0143±0.08%	0.0675±0.06%	**0.0116**±0.10%
SNR=20 dB	SAD	0.0376±0.16%	0.0176±0.42%	0.0394±2.92%	0.0145±0.53%	0.0176±0.42%	0.0176±0.42%	0.0405±0.56%	**0.0127**±0.73%
	RMSE	0.2201±0.62%	0.1603±0.58%	0.7257±5.32%	0.1661±0.85%	0.1625±2.05%	**0.1525**±0.45%	0.1610±0.81%	0.1626±0.93%
	RE	0.0934±0.19%	0.0873±0.25%	0.0856±0.53%	0.0895±0.47%	0.0994±0.37%	0.0977±0.30%	0.1547±0.55%	**0.0840**±0.42%
		676 Pixels							
		N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU
SNR=30 dB	SAD	0.0162±0.04%	0.0129±0.03%	0.0328±0.85%	0.0118±0.09%	0.0129±0.03%	0.0129±0.03%	0.0098±0.09%	**0.0113**±0.08%
	RMSE	0.4005±0.21%	0.3334±0.83%	0.9828±1.96%	0.3292±0.50%	0.4018±1.05%	**0.3157**±0.45%	0.3420±0.62%	0.3493±0.65%
	RE	0.0183±0.02%	0.0154±0.06%	0.0201±1.01%	0.0144±0.12%	0.0157±0.04%	**0.0136**±0.07%	0.0680±0.08%	0.0140±0.07%
SNR=20 dB	SAD	0.0410±0.10%	0.0197±0.15%	0.0435±1.30%	0.0193±0.61%	0.0197±0.24%	0.0197±0.20%	0.0428±0.73%	**0.0184**±0.52%
	RMSE	0.2335±0.58%	0.2416±0.23%	0.8876±2.09%	0.2846±1.03%	0.4469±1.27%	0.2514±0.74%	**0.2402**±0.91%	0.3046±1.08%
	RE	0.0997±0.21%	0.0856±0.16%	0.0883±0.92%	0.0924±0.84%	0.0939±0.45%	0.0998±0.70%	0.1548±0.68%	**0.0839**±0.39%

TABLE V

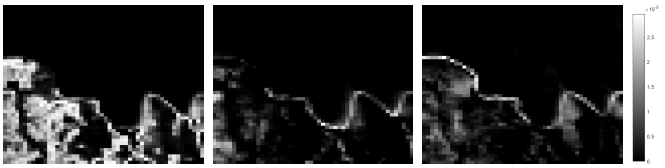
RES AND COMPUTATION TIME (IN SECONDS) OBTAINED BY DIFFERENT METHODS FOR THE MOFFETT FIELD DATA, WHERE THE BEST RESULTS ARE DISPLAYED IN BOLD

Attribute	LSMM-based				NSMM-based			
Method	N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU
RE	0.0210	0.0202	0.0280	0.0194	0.0192	0.0223	0.0207	**0.0187**
Time	0.9372	**0.3282**	0.3526	21.5662	123.1092	30.7614	40.3109	64.3542

50×50 pixels, with 189 bands covering the wavelength region from 0.4 to 2.5 μm (low SNR and water absorption bands were removed from the data). The selected region contains three endmembers, which are characteristic of the coastal area in the image: vegetation, water, and soil [70]. The subscene is widely used to evaluate algorithms for nonlinear unmixing in the remote sensing community [39].

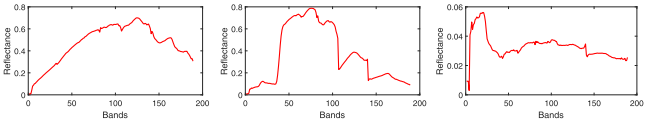
Due to the fact that Moffett data lack reference library signatures and truth abundances, our experiment just uses the RE to evaluate the accuracy of the obtained unmixing results. As shown in Table V, the proposed DMBU obtained the best RE value, although it takes a longer computation time. Moreover, the result also reveals that linear unmixing approaches generally exhibit lower computational complexity than nonlinear ones. For illustrative purposes, Fig. 6 displays the interaction abundance maps obtained by DMBU. Specifically, the scattering interaction between water and soil is mainly distributed in the boundaries between regions, and the scattering between soil and vegetation is quite high. Fig. 7 shows the endmembers estimated by DMBU, and Fig. 8 shows an abundance comparison among the nonlinear unmixing approaches. In Fig. 8, DMBU obtained competitive results when compared with other approaches. From this figure, we can conclude that the proposed DMBU can estimate the abundance distribution of the water better than the other tested methods.

Fig. 9 illustrates the scattering interactions estimated by different nonlinear approaches. The interaction distribution maps of rNMF and DMBU are obtained from the interaction outliers, while the ones of GBM and semi-NMF are obtained by multiplying the virtual endmembers and interaction abundances. To our knowledge, second-order scattering should be mainly present in the boundaries and mixed regions between



(a) (b) (c)

Fig. 6. Interaction abundance maps estimated by the DMBU in the Moffett Field data. (a) Soil-vegetation. (b) Vegetation-water. (c) Soil-water.



(a) (b) (c)

Fig. 7. Endmembers estimated by the proposed DMBU in the Moffett Field data set. (a) Soil. (b) Vegetation. (c) Water.

materials. Compared with the real distribution of materials in Fig. 9, we can observe that the scattering effects predicted by the proposed DMBU are reasonable, i.e., the interactions are mainly distributed in the boundaries and mixed regions between the materials. However, the other tested methods seem to find some problems. These results are consistent with our introspections on the expected distribution of nonlinear interactions.

2) Experiment With Jasper Ridge Data Set: The Jasper Ridge data (512×614 pixels) was gathered by AVIRIS over Jasper Ridge in California. The data consist of 224 bands over the wavelength region from 0.38 to 2.5 μm. Considering water vapor and atmospheric effects, bands 1–3, 108–112, 154–166, and 220–224 were removed, leaving a total of 198 spectral bands for our experiments [71]. To reduce the complexity

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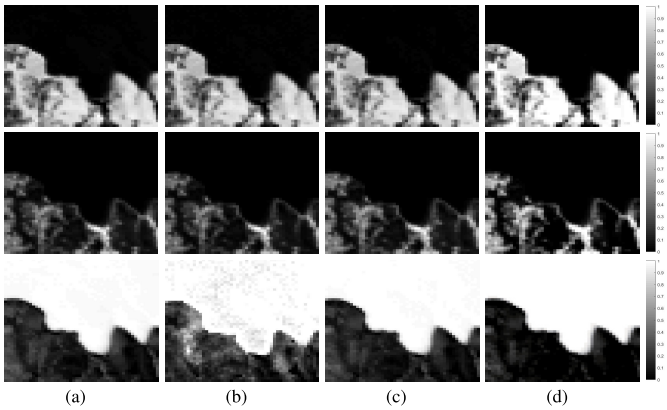


Fig. 8. Abundance maps estimated by nonlinear different methods in the Moffett Field data. (Top to bottom) Soil, vegetation, and water. (a) GBM. (b) Semi-NMF. (c) rNMF. (d) DMBU.

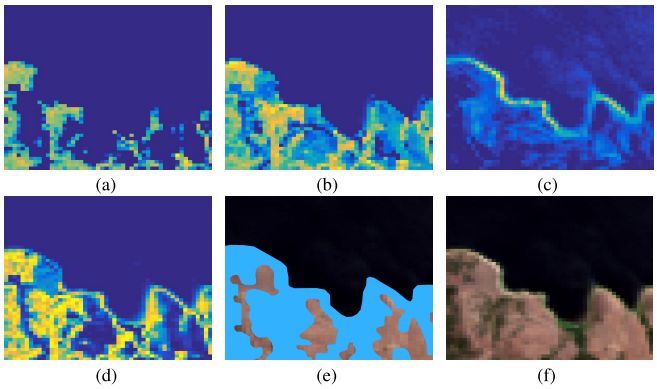


Fig. 9. Interaction distributions estimated by different nonlinear methods in the Moffett Field image. Bright yellow pixels: large values. Dark blue pixels: small ones. (a) GBM. (b) Semi-NMF. (c) rNMF. (d) DMBU. (e) Boundaries and mixing regions between materials, corresponding to the light blue area. (f) Real Moffett Field image.

of the data, a subimage with 100 × 100 pixels is selected from the original data set (this is a widely used image for testing unmixing algorithms). The subimage contains four endmembers: road, soil, tree, and water [71].

Table VI indicates that the proposed DMBU obtained the best RMSE and RE scores in this experiment. Moreover, the quantitative results illustrate that the proposed DMBU obtained better SADs. For illustrative purposes, the endmember signatures and abundance maps are shown in Fig. 10, respectively. Fig. 10 (middle) shows that the abundances estimated via DMBU, and it can be seen that these abundances match well with the corresponding references. Fig. 10 (bottom) reveals that the endmembers estimated via DMBU also exhibit a good match with the corresponding library signatures. The results in Table VI and Fig. 10 further demonstrate the effectiveness of the linear part of the proposed DMBU. On the other hand, Fig. 11(a)–(f) illustrates the interaction abundance maps estimated by DMBU in the Jasper Ridge data. Fig. 11(g) shows the interaction outliers obtained via the proposed approach. Compared with the real color Jasper Ridge data displayed in Fig. 11(h), most of the obtained results are reasonable, since the estimated interactions are distributed in the boundary and mixed regions between the constituent materials in the scene.

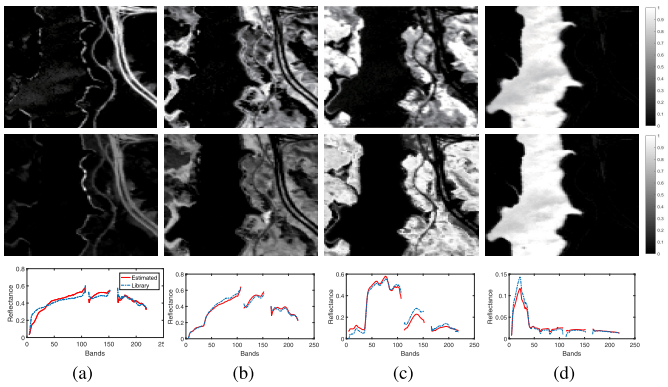


Fig. 10. Results obtained by DMBU on the Jasper Ridge data. (Top) Reference abundance maps. (Middle) Estimated abundance maps. (Bottom) Comparison of the estimated endmembers (red curves) with the reference signatures (blue curves). (a) Road. (b) Soil. (c) Tree. (d) Water.

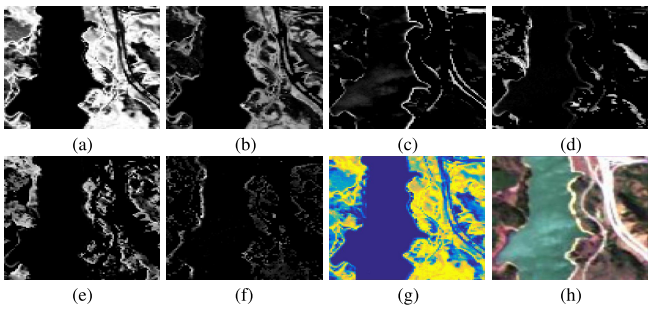


Fig. 11. Interaction abundance maps and interaction distribution map estimated by DMBU in the Jasper Ridge subimage. (a) Soil-tree. (b) Soil-water. (c) Soil-road. (d) Tree-road. (e) Tree-water. (f) Water-road. (g) Interaction distribution map. (h) Real image.

3) Experiment With Urban Data Set: The Urban data were acquired by the Hyperspectral Data Image Collection Experiment (HYDICE) instrument over the Copperas Cove, TX, USA, in 1995. [2] This region of the image contains a mixture of man-made objects and forestry. In the Urban data, the top of the scene contains a highway that crosses the region from left to right, a shopping mall along the highway, and a parking lot in front of the mall. The Urban data comprises a total of 307 × 307 pixels and 210 bands, with spectral resolution of 10 nm and spatial resolution of 2 × 2 m^2 per-pixel, covering the wavelength region from 0.4 to 2.5 μm [4], [72]. Prior to the analysis, bands 1–4, 76, 87, 101–111, 136–153, and 198–210 were removed due to water absorption and atmospheric effects in those bands, leaving a total of 162 spectral bands. In this experiment, the Urban data are used to further evaluate the proposed approach, and note that the scene includes six endmembers: asphalt, dirt, grass, metal, roof, and tree.

The quantitative results reported in Table VII reveal that the proposed DMBU obtained better results when compared with other methods in terms of the RMSE and RE values. For illustrative purposes, the estimated endmember signatures and abundances are shown in Fig. 12. In Fig. 12, we can observe that the abundance fractions and endmember signatures estimated by the proposed approach provide a very good match with regards to the corresponding ones in the ground-truth.

[2]https://www.agc.army.mil/Hypercube

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TABLE VI

SADs (IN RADIANS) RMSEs, AND REs OBTAINED BY DIFFERENT METHODS IN THE JASPER RIDGE DATA, WHERE THE BEST RESULTS ARE DISPLAYED IN BOLD

Method		N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU
SAD	Road	0.0906	0.1009	0.2560	0.1015	0.1009	0.1009	**0.0983**	0.0991
	Soil	0.2254	0.2093	**0.1908**	0.2179	0.2093	0.2093	0.1927	0.1920
	Tree	**0.1558**	0.1799	0.1844	0.1701	0.1799	0.1799	0.1730	0.1752
	Water	0.1337	0.1324	0.1584	0.1322	0.1324	0.1324	0.1326	**0.1319**
Mean SAD		0.1514	0.1556	0.1974	0.1554	0.1556	0.1556	**0.1492**	0.1496
RMSE		0.2754	0.2733	0.2641	0.2582	0.2736	0.2508	0.2479	**0.2474**
RE		0.0168	0.0170	0.0265	0.0166	0.0191	0.0196	0.0201	**0.0162**

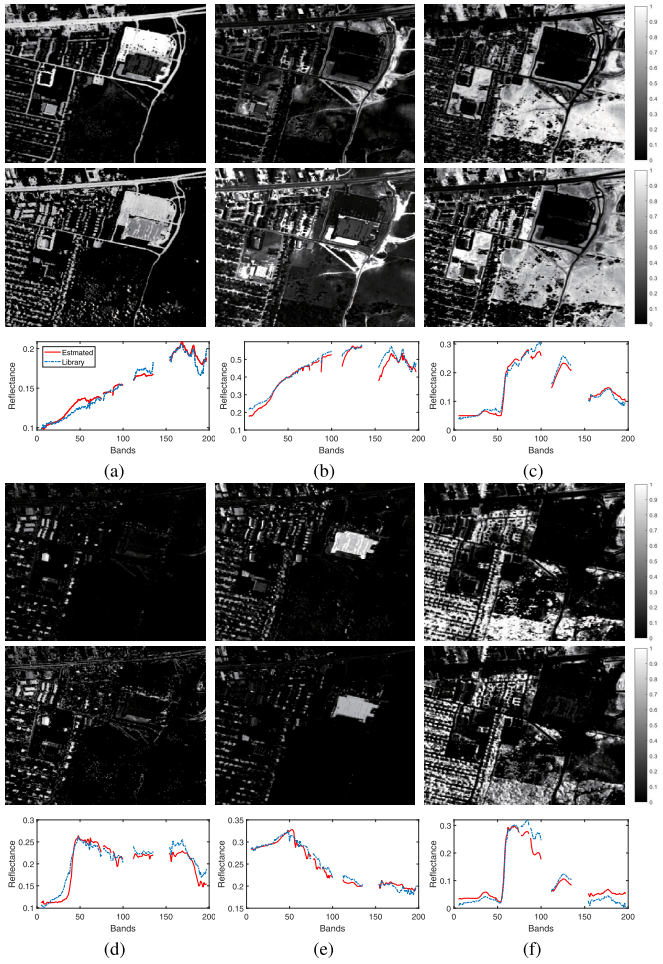


Fig. 12. Results estimated by the proposed DMBU in the Urban data. (Top) Reference abundance map. (Middle) Estimated abundance maps. (Bottom) Comparison of the estimated endmembers (red curves) with the reference signatures (blue curves). (a) Asphalt. (b) Dirt. (c) Grass. (d) Metal. (e) Roof. (f) Tree.

Fig. 13 displays the distribution map of interaction outliers which estimated by the DMBU. The figure reveals that the scattering interactions estimated by DMBU are reasonable, since they are distributed in the boundary regions between the constituent materials in the scene.

4) Experiment With Henry Island Data Set: The Henry Island data set was gathered by Earth-Observing-1 (EO-1) Hyperion and contains 137 × 187 pixels and 155 bands, with spatial resolution of 30 m. These data were obtained



Fig. 13. (a) Real Urban image. (b) Interaction distribution map estimated by the proposed DMBU.

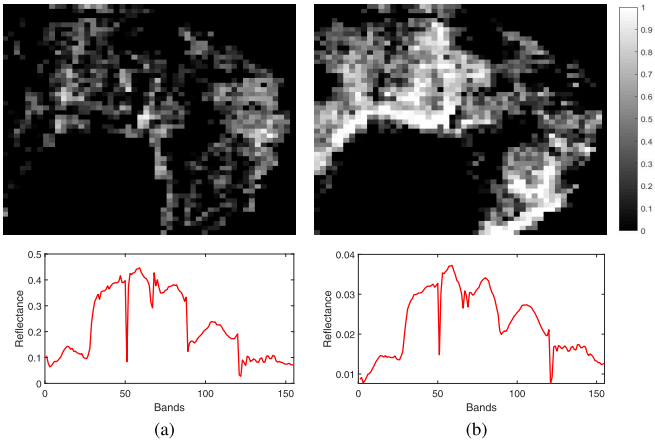


Fig. 14. Abundance maps and endmember signatures estimated by the proposed DMBU in the Henry Island data. (a) *Excoecaria Agallocha*. (b) *Ceriops Decandra*.

from the USGS Earth Resources Observation and Science Center through a data acquisition request to the satellite data provider. According to the description in [73], the atmospheric correction has converted the data to reflectance units by using the FLAASH model available in ENVI software. In this experiment, we select a subscene with 50×50 pixels from the Henry Island data. According to [73], the area in the subscene contains two endmembers: 1) *Excoecaria Agallocha* and 2) *Ceriops Decandra*.

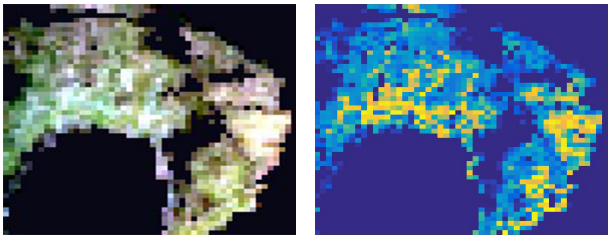
The abundances and endmembers estimated by the proposed DMBU are depicted in Fig. 14. Fig. 15 shows the distribution of interaction abundances and outliers, reflecting the distribution range of scattering interactions on the data. According to [73], *Excoecaria Agallocha* and *Ceriops Decandra* grow together in the field. As a result, the interactive distribution in Fig. 14 is reasonable, as it can also be seen in the real image.

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TABLE VII

SADs (IN RADIANS) RMSEs, AND REs OBTAINED BY DIFFERENT METHODS FOR THE URBAN DATA, WHERE THE BEST RESULTS ARE DISPLAYED IN BOLD

Method		N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU
SAD	Asphalt	0.2174	0.0491	0.1505	0.0532	0.0491	0.0491	0.0625	**0.0447**
	Dirt	0.1302	0.1089	0.1165	0.1043	0.1089	0.1089	**0.0990**	0.0993
	Grass	**0.0691**	0.1251	0.2395	0.1257	0.1251	0.1251	0.4164	0.1284
	Metal	0.8797	0.6921	**0.6807**	0.8265	0.6921	0.6921	0.7038	0.6825
	Roof	0.3644	**0.1002**	0.1077	0.1104	**0.1002**	**0.1002**	0.1763	0.1071
	Tree	0.4096	0.3547	0.3900	0.1895	0.4096	0.4096	**0.1754**	0.2418
Mean SAD		0.3451	0.2384	0.2808	0.2349	0.2384	0.2384	0.2722	**0.2173**
RMSE		0.3082	0.2294	0.2095	0.2106	0.2070	0.2083	0.2134	**0.2062**
RE		0.0302	0.0226	0.0254	0.0207	0.0219	0.0213	0.0206	**0.0201**



(a) (b)

Fig. 15. (a) Real Henry Island image. (b) Interaction distribution map estimated by the proposed DMBU.

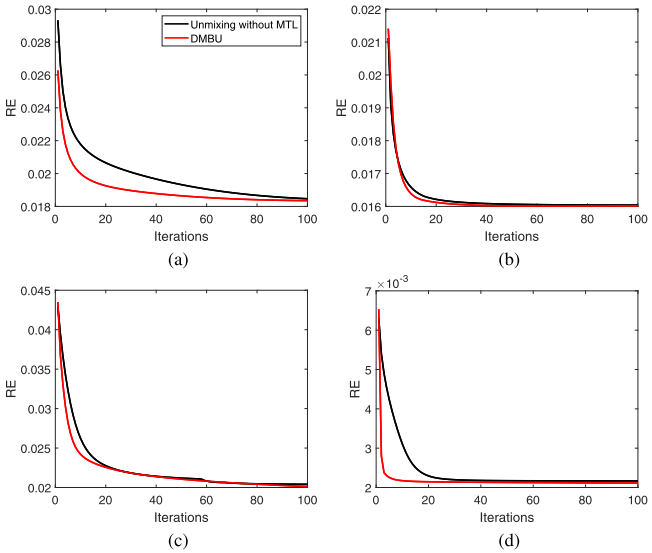


Fig. 16. Changes of the REs along with iterations, where these curves are obtained from different data sets. (a) Moffett Field. (b) Jasper Ridge. (c) Urban. (d) Henry Island.

In addition, after comparing the abundance map with the real image, we can observe that the results provided by our DMBU exhibit a good match with the distributions of the two plants in the real scene. These results further verify the effectiveness of our method.

5) Comparison on Optimization: The proposed DMBU adopts the MTL, aiming at handling the scattering interactions when estimating endmembers and abundances. It is worth noting, however, that the optimization performance may be different between the unmixing method with and without the MTL framework. For the proposed unmixing approach, the convergence on RE can reflect the optimization performance. The changes of the REs along with iterations are illustrated in Fig. 16, where the results are obtained on Moffett Field, Jasper Ridge, Urban, and Henry Island data sets, respectively. Concerning the two tasks in DMBU, task 1 can be performed individually, while task 2 cannot be implemented alone because it needs to be activated via the nodes of the hidden layers in task 1. In this regard, the proposed approach without the MTL framework only performs task 1. In Fig. 16, the red curves represent the results of DMBU, while the black ones are obtained for the proposed method without the MTL framework. To ensure a fair comparison, the number of iterations of the proposed methods is set to 100. From the curves shown in Fig. 16, we can observe that the optimizations are successful because these REs can reach convergence. Note that the unmixing method without the MTL framework only obtains endmembers and abundances, and the corresponding REs only relate to the linear components.

V. Conclusion and Future Lines

Autoencoders have successfully been used to solve unmixing problems by focusing on the LMM. Moreover, most of the previous research works based on autoencoders only focused on increasing the depth of the networks to improve unmixing performance. However, our introspection in this article is that simply increasing the depth of the networks may not be sufficient to model second-order scattering. To address this issue, in this article we develop a new method that combines two deep autoencoders with an MTL framework to conduct bilinear unmixing. Our framework, called DMBU, models the unmixing problem by using two different tasks that are fine-tuned by an iterative scheme. Our experimental results indicate that the newly developed DMBU is able to accurately estimate endmember signatures and abundance factions, while also modeling the impact of second-order scattering interactions. In order to enhance the practical exploitation of our proposed approach, we will consider in further detail the spatial correlation and spectral variability present in real scenarios.

Appendix

In this section, we provide the detailed derivation steps of the partial derivatives of $\mathcal{C}_{\text{Task1}}$. The partial derivatives of $\mathcal{C}_{\text{Task1}}$

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with respect to $\mathbf{A}_k$ is given as

$$\begin{aligned}
&\nabla_{\mathbf{A}_k}\mathcal{C}_{\text{Task 1}} \\
&= \frac{1}{2}\frac{\partial\|\mathbf{Y}-\mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_m\mathbf{A}_m\|_F^2}{\partial\mathbf{A}_k}+\frac{1}{2}\mu\frac{\partial\Re(\mathbf{A}_k)}{\partial\mathbf{A}_k} \\
&= \frac{1}{2}\partial\,\mathrm{Tr}(\mathbf{Y}^\top\mathbf{Y}-2\mathbf{Y}^\top\mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_m\mathbf{A}_m+\mathbf{A}_m^\top\mathbf{E}_m^\top\mathbf{E}_{m-1}^\top \\
&\quad\cdots\mathbf{E}_1^\top\mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_m\mathbf{A}_m)/\partial\mathbf{A}_k+\frac{1}{2}\mu\frac{\partial\,\mathrm{Tr}\left(\mathbf{A}_k\mathbf{L}\mathbf{A}_k^\top\right)}{\partial\mathbf{A}_k} \\
&= \frac{1}{2}\frac{\partial\,\mathrm{Tr}\left(\mathbf{Y}^\top\mathbf{Y}-2\mathbf{Y}^\top\boldsymbol{\Psi}\mathbf{E}_k\mathbf{A}_k+(\boldsymbol{\Psi}\mathbf{E}_k\mathbf{A}_k)^\top\boldsymbol{\Psi}\mathbf{E}_k\mathbf{A}_k\right)}{\partial\mathbf{A}_k} \\
&\quad+\frac{1}{2}\mu(\mathbf{A}_k\mathbf{L}+\mathbf{A}_k\mathbf{L}^\top) \\
&= \mathbf{E}_k^\top\boldsymbol{\Psi}^\top\boldsymbol{\Psi}\mathbf{E}_k\mathbf{A}_k-\mathbf{E}_k^\top\boldsymbol{\Psi}^\top\mathbf{Y}+\frac{1}{2}\mu(\mathbf{L}+\mathbf{L}^\top)\mathbf{A}_k \\
&= \mathbf{E}_k^\top\boldsymbol{\Psi}^\top(\boldsymbol{\Psi}\mathbf{E}_k\mathbf{A}_k-\mathbf{Y})+\mu\mathbf{A}_k\mathbf{L}. \qquad (29)
\end{aligned}$$

Similarly, the partial derivative of $\mathcal{C}_{Task1}$ with respect to $\mathbf{E}_k$ is obtained as

$$\begin{aligned}
&\nabla_{\mathbf{E}_k}\mathcal{C}_{\text{Task 1}} \\
&= \frac{1}{2}\frac{\partial\,\mathrm{Tr}\left(\mathbf{Y}^\top\mathbf{Y}-2\mathbf{Y}^\top\boldsymbol{\Psi}\mathbf{E}_k\widetilde{\mathbf{A}}_k+(\boldsymbol{\Psi}\mathbf{E}_k\widetilde{\mathbf{A}}_k)^\top\boldsymbol{\Psi}\mathbf{E}_k\widetilde{\mathbf{A}}_k\right)}{\partial\mathbf{E}_k} \\
&= \boldsymbol{\Psi}^\top\boldsymbol{\Psi}\mathbf{E}_k\widetilde{\mathbf{A}}_k\widetilde{\mathbf{A}}_k^\top-\boldsymbol{\Psi}^\top\mathbf{Y}\widetilde{\mathbf{A}}_k^\top \\
&= \boldsymbol{\Psi}^\top(\boldsymbol{\Psi}\mathbf{E}_k\widetilde{\mathbf{A}}_k-\mathbf{Y})\widetilde{\mathbf{A}}_k^\top. \qquad (30)
\end{aligned}$$

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